

Module 9: Systems of Linear Equations – Prequiz

Name:

Date:

Four short questions. Interpret a solution as a common point satisfying every equation.

1. Solve the system $x + y = 8$ and $x - y = 2$.
2. Classify the system $2x + y = 4$ and $4x + 2y = 10$ as having one solution, no solution, or infinitely many solutions. Explain briefly.
3. Solve the triangular system
$$z = 2, \quad y - z = 1, \quad x + y + z = 9.$$
4. What is the geometric meaning of a unique solution to two linear equations in two variables?

Module 9: Systems of Linear Equations – Weekly Check-In

Name:

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Three questions. Interpret a solution as a common point satisfying every equation.

1. Solve and interpret a two-variable system.

Solve

$$\begin{cases} 2x + y = 7, \\ x - y = 2. \end{cases}$$

(a) Solve algebraically and state the ordered-pair solution.

(b) Verify the ordered pair in both original equations.

(c) What does this solution represent geometrically?

2. Classifying systems.

Classify each system as having one solution, no solution, or infinitely many solutions. Explain using the relationship between the equations.

(a) $2x + 3y = 6$ and $4x + 6y = 12$.

(b) $2x + 3y = 6$ and $4x + 6y = 10$.

Module 9: Systems of Linear Equations – Weekly Check-In, continued**Name:****Date:****3. Back-substitution in three variables.**

Solve

$$\begin{cases} z = 2, \\ y - 3z = -1, \\ 2x + y + z = 15. \end{cases}$$

Show the order in which you solve for the variables. Then check your ordered triple in all three equations.